

National Academy of Science and Technology

(Affiliated to Pokhara University)

Dhangadhi, Kailali

Pre-University Examination

Semester : V_Fall

Level: Bachelor

Program: B.E. Computer

Course: Embedded System

Year : 2025

F.M. : 100

P. M. : 45

Time : 3 Hrs.

Candidates are required to give their answer in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Compare Embedded systems with general purpose computing systems in terms of design goals and constraints. 7
b) Describe the AVR Architecture. Write an embedded C program to read data from PORT B and, if the value is greater than 100 send it to PORT C, otherwise clear PORT C. 8
2. a) Discuss the various Serial Communication features available in AVR microcontrollers. 8
b) Explain the functionality of counters in AVR. How can you configure a timer to count external pulses? Assume necessary register details. 7
3. a) Define Task Scheduling and context switching in the context of a Real Time Operating System(RTOS). 8
b) What is priority Inversion? Explain how it can lead to system failure and discuss one method to prevent it. 7
4. a) Mention the features of VHDL and the advantages associated with them. 8
b) Write a VHDL test bench program for full adder. 7
5. a) Write a VHDL program to detect a sequence '1011'. 7
b) Write a C program for ATmega32 to display "Hello World" in a 16*2 LCD (4-bit mode). 8
6. a) Compare the features and typical use cases of Bluetooth, ZigBee and LoRa wireless protocols. 8
b) Write about MQTT protocol used in IoT Communication. 7
7. Write short notes on any two: 2x5
 - a) Raspberry Pi
 - b) Deadlock in RTOS
 - c) Sensor